

Probability and Distributions

STA257H1 - Probability and Statistics I

Omidali A. Jazi

University of Toronto
Department of Statistical Sciences

Summer 2026

Goals for this Chapter

- Introduction
- Sets
- The Probability Set Function
- Conditional Probability and Independence
- Random Variables
- Discrete Random Variables
- Continuous Random Variables
- Expectation of a Random Variable
- Some Special Expectations
- Important Inequalities

Section 1

Introduction

Defining Probability

- In every day conversation, probability is a measure of a person's belief in the occurrence of a future event.

- **Example:** What is the chance of winning an election?

In this sense, probability is completely **subjective**.

- More quantitatively, we define probability as the fraction of times an event occurs if it's repeated over and over (infinitely).

This is the **relative frequency** definition of probability.

Probability vs Statistics

- What is the difference between probability and statistics?
- In probability, the **randomization mechanism** is assumed/known and calculations follow.

Example: Given a fair coin, what's the chance of observing 8 heads out of 10 coin flips?

- In statistics, the outcome is known and used to infer something about the mechanism.

Example: Given that I have just observed 8 heads out of 10 coin flips, is the coin fair?

Probability Quantifies Outcome Uncertainty

- Imagine someone is flipping a coin. With enough practice, we can learn to flip a coin and have it come up heads every time.

In this case, there is no uncertainty in the outcome. But without practice, the coin is going to come up heads and tails "randomly".

- There is randomness because the particular outcomes are uncontrollable and/or unpredictable.
- Probability helps you quantify the likelihood of various outcomes.

Probability and Inference

- In statistics, a sample of data is used to say something about the entire population.
- We learn formal methods for making inference such as hypothesis testing and model fitting.
- But first, we need to understand the theory of probability. Both what it is and, perhaps more importantly, how to calculate it.

Section 2

Sets

Terminologies and Notation

To begin, we need some terminologies and notation from the set theory.

- Capital letters denote the sets of objects: A , B , etc.
- If set A consists of objects a_1, a_2, a_3 , we write $A = \{a_1, a_2, a_3\}$
- S denotes the **universal set** which is the set of all possible objects.
- ϕ denotes the **null** or **empty set** ($\phi = \{\}$) which is the set without any object.

Terminologies and Notation

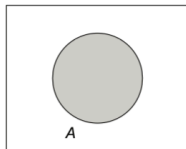
- For any two sets A and B , A is a **subset** of B if every object in A is in B . Notation: $A \subset B$.
- **Example:** If $B = \{a_1, a_2, a_3, a_4\}$ and $A = \{a_1, a_2, a_3\}$ then A is a subset of B .
- Note: The null set is a subset of every set.
- The **union** of A and B is the set of all objects in A or B or both. Notation: $A \cup B$.
- **Example:** If $A = \{a_1, a_2\}$ and $B = \{b_1, b_2\}$ then $A \cup B = \{a_1, a_2, b_1, b_2\}$.

Terminologies and Notation

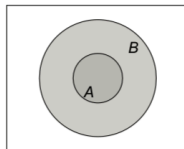
- The **intersection** of sets A and B is the set of all objects in both A and B . Notation: $A \cap B$ or just AB .
- **Example:** If $B = \{a_1, a_2, a_3, a_4\}$ and $A = \{a_1, a_2, a_3\}$ then $A \cap B = \{a_1, a_2, a_3\}$.
- The key word for union is or while the key word for intersection is and.

Venn Diagrams

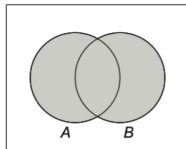
Panel (a)

 A

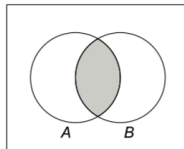
Panel (b)

 $A \subset B$

Panel (c)

 $A \cup B$

Panel (d)

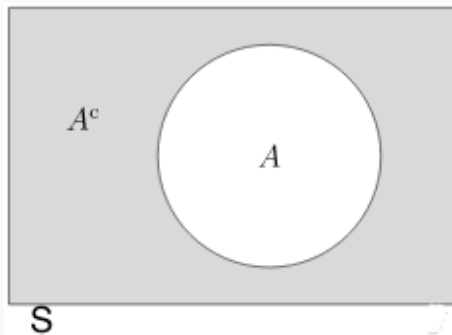
 $A \cap B$

Terminologies and Notation

- If A is a subset of S , then the **complement** of A is denoted by A^c .
- The complement of A is the set of all objects in S that are not in A .
- Note that $A \cup A^c = S$
- Two sets A and B are said to be **disjoint** or **mutually exclusive** if they have no object in common. That is, A and B are disjoint if $A \cap B = \phi$

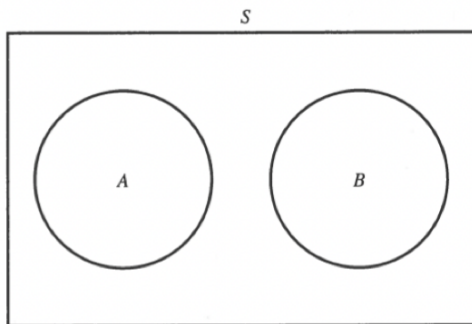
Note: By definition, $A \cap A^c = \phi$

Venn Diagram: Complement



Venn Diagram: Mutually Exclusive

Venn diagram for
mutually exclusive
sets A and B



Laws

- Distributive Laws:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- DeMorgan's Laws:

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

Nondecreasing & Nonincreasing Sets

A sequence of sets $\{C_n\}$ is nondecreasing if $C_n \subset C_{n+1}$, for all n , in which we write

$$\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n.$$

A sequence of sets $\{C_n\}$ is nonincreasing if $C_{n+1} \subset C_n$, for all n , in which we write

$$\lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n.$$

Example 1: Show that the sequence of sets $C_k = \{x : \frac{1}{k} \leq x \leq 3 - \frac{1}{k}\}$ for $k = 1, 2, 3, \dots$ is increasing and find $\lim_{k \rightarrow \infty} C_k$.

Example 2: Show that the sequence of sets $C_k = \{x : 2 < x \leq 2 + \frac{1}{k}\}$ for $k = 1, 2, 3, \dots$ is decreasing and find $\lim_{k \rightarrow \infty} C_k$.

Section 3

The Probability Set Function

Random Experiment, Event, Sample Space

Definition 1

A **random experiment** is an act or process of observation that leads to a single outcome which cannot be predicted with certainty.

Definition 2

The **sample space** of an experiment is the set of all possible outcomes (sample points) and is denoted by S .

Definition 3

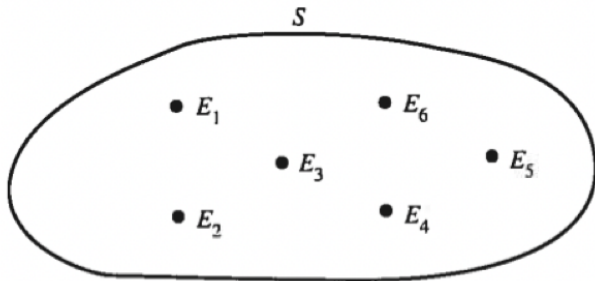
A **simple event** is an event that cannot be decomposed. Each simple event corresponds to one and only one **sample point** which is the most basic outcome of a random experiment. A **compound event** consists of two or more simple events.

Example

- In the six-sided die tossing experiment,
 $S = \{E_1, E_2, E_3, E_4, E_5, E_6\}$ where E_i represents "observe number i ".
- The letter E with a subscript will be used to denote a simple event or the corresponding sample point.
- The events "observe a 1," "observe a 3" and "observe a 5" are all simple events since they cannot be decomposed any further. Each simple event corresponds to only one sample point.
- However, the event "observe an odd outcome" on the six-sided die experiment is a compound event since it is composed of three simple events: "observe a 1," "observe a 3," and "observe a 5".

Example

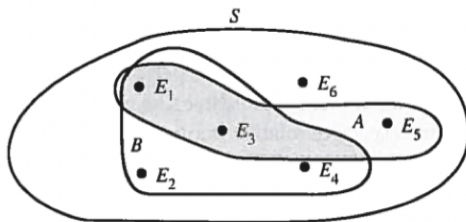
Venn diagram for the
sample space
associated with
the die-tossing
experiment



Example

The compound event A “observe an odd number” is $A = \{E_1, E_3, E_5\}$. We could also write $A = \{E_1\} \cup \{E_3\} \cup \{E_5\}$. Define event B to be “observe a number less than 5” or $B = \{E_1, E_2, E_3, E_4\}$.

Venn diagram for the die-tossing experiment



Discrete Sample Space & Event

Definition 4

A **discrete sample space** is one that contains countable (finite or infinite) number of distinct sample points.

Definition 5

An **event** in a discrete sample space S is a collection of sample points. That is, any subset of S .

Example: A random experiment that involves tossing three dice at the same time until the same numbers come up.

Probability Axioms

On analyzing the relative frequency concept of probability, we see that three conditions must hold.

- ❶ The relative frequency of any event must be greater than or equal to zero. Hence, negative frequency does not make sense.
- ❷ The relative frequency of the whole sample space S must be 1 because all possible experimental outcomes are in S , something in S must occur every time the experiment is conducted.
- ❸ If two events are mutually exclusive, then the probability of their union is the sum of their respective frequencies.

Probability Axioms

Definition 6

Suppose S is a sample space. To every event A in S , we assign a number $P(A)$, called the probability of A , so that the following axioms hold:

- Axiom 1: $P(A) \geq 0$
- Axiom 2: $P(S) = 1$
- Axiom 3: If $A_1, A_2, A_3, \dots$ are a sequence of **pairwise mutually exclusive** events in S (i.e, $A_i \cap A_j = \phi$ if $i \neq j$)

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

The Addition and Complement Rules

Theorem 7

(The Addition Law of Probability): The probability of the union of two events A and B is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

Theorem 8

For an event A ,

$$P(A) = 1 - P(A^c).$$

Proof: In class.

Example 1: Gene Expression Profiling

Gene expression profiling is a state-of-the-art method for determining the biology of cells. In Briefings in Functional Genomics and Proteomics (Dec. 2006), biologists at Pacific Northwest National Laboratory reviewed several gene expression profiling methods. The biologists applied two of the methods (A and B) to data collected on proteins in human mammary cells.

The probability that the protein is cross-referenced (i.e., identified) by method A is 0.41, the probability that the protein is cross-referenced by method B is 0.42, and the probability that the protein is cross-referenced by both methods is 0.40.

Example 1: Gene Expression Profiling

- Find the probability that the protein is cross-referenced by either method A or method B .

Solution: $0.41 + 0.42 - 0.40 = 0.43$

- On the basis of your answer to part a, find the probability that the protein is not cross-referenced by either method.

Solution: $1 - 0.43 = 0.57$

{Calculating Probabilities: Sample-Point Method}

- ➊ First, define the experiment and clearly determine how to describe one simple event.
- ➋ List the simple events and test to make sure they cannot be decomposed. This defines S .
- ➌ Assign reasonable probabilities to the sample points in S making sure $P(A_i) \geq 0$ and $P(\cup_i A_i) = 1$.
- ➍ Define the event of interest A as a specific collection of sample points.
- ➎ Find $P(A)$ by summing the probabilities of the sample points in A .

Example

Consider a fair six-sided die. Let A_i ($i = 1, 2, \dots, 6$) denote the event “observe number i ”. Then, by Axioms 2 and 3:

$$\begin{aligned} P(S) &= P(A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6) \\ &= P(A_1) + P(A_2) + P(A_3) + P(A_4) + P(A_5) + P(A_6) \\ &= 1 \end{aligned}$$

Counting Sample Points

- Let's look at some useful and powerful ways to count numbers of events.
- A sample space contains N equally likely events. An event A contains exactly n_a sample points, then we have

$$P(A) = \frac{n_a}{N}$$

- So, if we know N and can count n_a , then we can calculate the probability of event A .

Pros & Cons of Sample-Point Method

- Sample-point approach is direct and powerful, but prone to human error.
- Mistakes include mis-defining simple events and missing simple events. For sample, spaces with large number of events, can be challenging (perhaps impossible) to count all the events.
- This leads us to learn about combinatorial analysis (combinations and permutations).

The mn Rule and Extension

Theorem 9

With m elements $\{a_1, a_2, \dots, a_m\}$ and n elements $\{b_1, b_2, \dots, b_n\}$, it is possible to form $m \times n$ pairs containing one element from each group.

Extension: If we have k sets of elements, n_1 in the first set, n_2 in the second set, ..., and n_k in the k th set. Suppose you wish to form a sample of k elements by taking one element from each of the k sets. Then the number of different samples that can be formed is $n_1 n_2 \dots n_k$.

The rule is also known as **Multiplication Rule**.

Permutations (Counting Ordered Arrangements)

Definition 10

An **ordered** arrangement of r distinct objects is called a **permutation**. The number of ways of ordering n distinct objects taken r at a time is denoted by P_r^n .

Theorem 11

To calculate the number of ways r items can be ordered out of n

$$P_r^n = n(n-1)(n-2) \cdots (n-r+1) = \frac{n!}{(n-r)!}$$

Combinations (Counting Unordered Arrangements)

Definition 12

The number of **unordered** subsets of r objects out of n is called the number of **combinations**. That is, it is the number of subsets, of size r , that can be formed from n objects. The notation is C_r^n .

Theorem 13

The number of unordered subsets of size r chosen (without replacement) from n objects is

$$C_r^n = \binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$$

The term $\binom{n}{r}$ is generally referred as the binomial coefficient because they occur in the binomial expansion $(x + y)^n$.

Combinations: Extension

Theorem 14

The number of ways of partitioning n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects, respectively, where each object appears in exactly one group and $\sum_{i=1}^k n_i = n$, is

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

*The terms $\binom{n}{n_1, n_2, \dots, n_k}$ are called the **multinomial coefficients** because they occur in the expansion of the multinomial term $(y_1 + y_2 + \dots + y_k)^n$.*

Examples

- We have $n = 3$ people (A , B , and C) and $r = 2$ job positions. How many ways (permutations) are there to assign the positions to the people?

Solution: There are 6 orderings: $\{A, B\}$, $\{A, C\}$, $\{B, A\}$, $\{B, C\}$, $\{C, A\}$, $\{C, B\}$

- We have $n = 3$ people (A , B , and C) and pick $r = 2$ people to create a committee. How many combination of two types of three?

Solution: There are only three: $\{A, B\}$, $\{A, C\}$, $\{B, C\}$.

- You need to assign 9 aircrafts to 3 sorties of size 4, 3, and 2 aircrafts. How many ways can the 9 aircrafts be assigned to the 3 groups?

Solution: $\frac{9!}{4!3!2!}$

Additional Properties of Probability

Inclusion-Exclusion Formula: The Addition Law of Probability can be extended to k events, $A_1, A_2, \dots, A_k$ as

$$\begin{aligned} P\left(\bigcup_{i=1}^k A_i\right) \\ = \sum_{i=1}^k P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \dots + (-1)^{k-1} P(A_1 \cap \dots \cap A_k) \end{aligned}$$

Proof: Exercise! (use induction)

Additional Properties of Probability

Theorem 15

Let $\{C_n\}$ be a increasing sequence of events. Then

$$\lim_{n \rightarrow \infty} P(C_n) = P\left(\lim_{n \rightarrow \infty} C_n\right) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$$

Let $\{C_n\}$ be a decreasing sequence of events. Then

$$\lim_{n \rightarrow \infty} P(C_n) = P\left(\lim_{n \rightarrow \infty} C_n\right) = P\left(\bigcap_{n=1}^{\infty} C_n\right)$$

Proof: In class.

This theorem is referred to as the **Continuity Theorem of Probability**.

Additional Properties of Probability

Another useful result for arbitrary unions is given by

Theorem 16

(Boole's Inequality) Let $\{C_n\}$ be an arbitrary sequence of events. Then

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) \leq \sum_{n=1}^{\infty} P(C_n)$$

Proof: In class.

Hint: Consider $D_1 = C_1$ and $D_n = C_n - \bigcup_{i=1}^{n-1} C_i$ for $n = 2, 3, \dots$. Then

$\{D_n\}$ is a disjoint sequence of events, and further $\bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} D_n$.

Now, use Axiom 3 to prove the theorem.

Additional Properties of Probability

From the Addition Law of Probability, we have

$$P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2) \leq 1$$

which gives **Bonferroni's Inequality** as

$$P(C_1 \cap C_2) \geq P(C_1) + P(C_2) - 1$$

Section 4

Conditional Probability and Independence

Conditional Probability

- The idea of conditional probability is that the probability of an event depends on our knowledge of other events that have occurred.
- **Example:** Imagine I am holding a card I picked at random out of a standard deck of playing. What is the probability that it is the ace of hearts?
- **Answer:** It's $\frac{1}{52}$ or just under 0.02.
- Now, if I tell you the card I am holding is an ace, what is the chance I am holding the ace of hearts?
- **Answer:** It's $\frac{1}{4}$ since we know it has to be one of only 4 aces in the deck.

Conditional Probability

Definition 17

The **conditional probability** of an event A , given that an event B has occurred, is equal to

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

provided that $P(B) > 0$.

Back to the Example: Ace of Hearts

- Define the events: A , choose the ace of hearts, and B , choose an ace (randomly out of 52 cards).
- Calculate the probabilities: Let's start with $P(B)$.
- Sample space S consists of 52 sample points (distinct cards), all of which are equally likely to be chosen.
- $P(B) = \frac{4}{52}$ (there are 4 aces in 52 cards).
- Now, $P(A \cap B)$ means "the probability of choosing the ace of hearts and choosing an ace".
- In this case, that is the same as "the probability of choosing the ace of hearts"
- So, $P(A \cap B) = P(A) = \frac{1}{52}$. Then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/52}{4/52} = \frac{1}{4}$$

Multiplication Law of Probability

Theorem 18

The probability of the intersection of two events A and B is

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

Proof: In class.

Multiplication Law of Probability: Extension

By twice applying the theorem

$$\begin{aligned}P(A \cap B \cap C) &= P[(A \cap B) \cap C] = P(A \cap B)P(C|A \cap B) \\&= P(A)P(B|A)P(C|A \cap B)\end{aligned}$$

And, this can be extended to any number, say k , events:

$$\begin{aligned}P(A_1 \cap A_2 \cap \dots \cap A_k) &= \\&P(A_1)P(A_2|A_1) \dots P(A_k|A_1 \cap A_2 \cap \dots \cap A_{k-1})\end{aligned}$$

The Multiplication Law is very useful for calculating complicated probabilities when conditional probabilities are easier to determine.

Example

A university consists of 60% female and 40% male students. If a student is female, 70% and 30% are the chances of being local and international respectively. Further, half of of the female local students are undergraduate, while 30% and 20% are master and PhD, respectively.

If a student is randomly chosen, find the probability of choosing a female local undergraduate student.

Solution: $0.6 \times 0.7 \times 0.5 = 0.21$

Law of Total Probability

It is often useful to view a sample space as the union of mutually exclusive subsets.

Definition 19

For some positive integer k , let the sets $\{B_1, B_2, \dots, B_k\}$ be such that

❶ $S = B_1 \cup B_2 \cup \dots \cup B_k$

❷ $B_i \cap B_j = \phi$ for $i \neq j$.

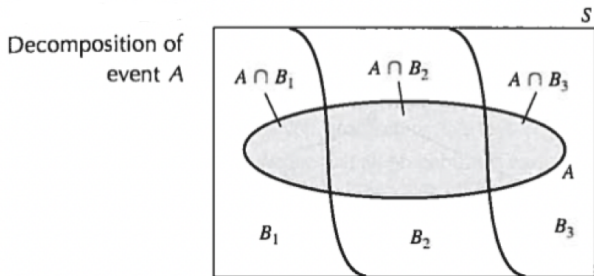
Then the collection of sets $\{B_1, B_2, \dots, B_k\}$ is said to be a **partition** of S .

Using the Partition to Decompose a Set

Now, if A is any subset of S and $\{B_1, B_2, \dots, B_k\}$ is a partition of S , then A can be decomposed as follows:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_k)$$

An illustration for $k = 3$



Law of Total Probability

Theorem 20

Assume that $\{B_1, B_2, \dots, B_k\}$ is a partition of S such that $P(B_i) > 0$ for $i = 1, \dots, k$. Then for any event A ,

$$P(A) = \sum_{i=1}^k P(A|B_i)P(B_i)$$

Proof: In class.

Example

Of the voters in a city, 40% are Republican and 60% are Democrats. Among the Republicans, 70% are in favor of a bond issue, while 80% of the Democrats favor this issue.

If a voter is selected at random, what is the probability that he/she will favor the bond issue?

Solution: $P(F) = 0.7 \times 0.4 + 0.8 \times 0.6 = 0.76$

Bayes' Theorem

Theorem 21

Assume that $\{B_1, B_2, \dots, B_k\}$ is a partition of S such that $P(B_i) > 0$ for $i = 1, \dots, k$. Then,

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^k P(A|B_i)P(B_i)}$$

Proof: In class.

Applications of the Bayes' Theorem

- Invert conditional probabilities: You know the various conditional probabilities $P(A|B_1)$, $P(A|B_2), \dots, P(A|B_k)$ but you want to know $P(B_j|A)$ for some B_j
- You know/obtain $P(B_1)$, $P(B_2), \dots, P(B_k)$ and then subsequently observe/collect some information $P(A|B_1)$, $P(A|B_2), \dots, P(A|B_k)$.
- Update your probabilities conditioned on what you have observed by calculating $P(B_1|A)$, $P(B_2|A)$, $\dots, P(B_k|A)$.

Example: Drug Testing

Based on an investigation, it is believed that 1% of the sailors use illegal drugs. A test detects drug use as follows:

- If someone has used illegal drugs, there is a 99% chance it will detect it (true positive).
- If someone does not use illegal drugs, there is a 0.5% chance the test will indicate it (false positive).

What is the chance that a positive drug test signals a drug user?

Example: Drug Testing

Solution:

- U_+ is the event an individual used illegal drugs, U_- is the event he/she did not.
- T_+ is the event the drug test is positive for the individual.
- We know $P(T_+|U_+) = 0.99$ and we want to calculate $P(U_+|T_+)$.

$$\begin{aligned} P(U_+|T_+) &= \frac{P(T_+|U_+)P(U_+)}{P(T_+|U_+)P(U_+) + P(T_+|U_-)P(U_-)} \\ &= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.005 \times (1 - 0.01)} \\ &= \frac{2}{3} \end{aligned}$$

Exercise: Maize Seeds

The genetic origin and properties of maize (modern-day corn) were investigated in Economic Botany. Seeds from maize ears carry either single spikelets or paired spikelets, but not both. Progeny tests on approximately 600 maize ears revealed the following information:

40% percent of all seeds carry single spikelets, while 60% carry paired spikelets. A seed with single spikelets will produce maize ears with single spikelets 29% of the time and paired spikelets 71% of the time. A seed with paired spikelets will produce maize ears with single spikelets 26% of the time and paired spikelets 74% of the time.

Find the probability that a randomly selected maize ear seed carries a single spikelet and produces ears with single spikelets.

Find the probability that a randomly selected maize ear seed produces ears with paired spikelets.

Independence

- The idea of independent events is that the probability of occurring event A is unaffected by whether B occurs or not.
- Another way to think about it is that knowledge about event B does not affect your probabilistic assessment of A . This is the intuitive notion that A and B are independent.
- But, if knowledge of B tells you something about the probability of A , then the two events are dependent.

Independent Events

Definition 22

Two events A and B are said to be **independent** if any one of the following holds:

- $P(A|B) = P(A)$
- $P(B|A) = P(B)$
- $P(A \cap B) = P(A)P(B)$

Otherwise, the events are dependent.

Note that each condition follows from either of the other two conditions and the definition of conditional probability.

Example 1

It is known that a patient with a disease will respond to treatment with probability 0.9. If three patients with the disease are treated and they respond independently, find the probability that at least one will respond.

Solution: Let A_i , $i = 1, 2, 3$ be the event that patient i responds to the treatment. Then, by Theorem 8, DeMorgan's Law, and the definition of independence, we have

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= 1 - P(A_1^c \cap A_2^c \cap A_3^c) \\ &= 1 - P(A_1^c)P(A_2^c)P(A_3^c) \\ &= 1 - 0.1^3 = 0.999 \end{aligned}$$

Exmample 2

Consider the following events in the toss of a single balanced six-sided die:

- A = Observe an odd number
- B = Observe an even number
- C = Observe a 1 or a 2.
- Question 1: Are A and B independent events?

Solution: No, because

$$P(A \cap B) = 0 \neq P(A)P(B) = \frac{3}{6} \times \frac{3}{6} = \frac{1}{4}$$

Conclusion: Two disjoint (mutually exclusive) events with positive probabilities are always dependent.

Example 2

- Question 2: Are A and C independent events?

Solution: Yes, because

$$P(A \cap C) = \frac{1}{6} = P(A)P(C) = \frac{3}{6} \times \frac{2}{6} = \frac{1}{6}$$

Conclusion: Disjoint events and independent events are different notions and should not be mixed up. In this case, two events share a sample point, but they are independent.

Section 5

Random Variables

Random Variables

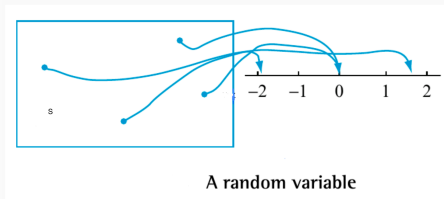
- Next, we will introduce random variables, probability and distribution functions, and the development of the concepts necessary for carrying out calculations for a probability model using these entities.
- The concept of a random variable allows us to pass from the experimental outcomes themselves to a numerical function of the outcomes.

Random Variables

Definition 23

Consider a random experiment with a sample space $\mathcal{C}$. A function X , which assigns to each element $c \in \mathcal{C}$ one and only one number $X(c) = x$, is called **random variable**. The space or range of X is the set of real numbers $\mathcal{D} = \{x : x = X(c), c \in \mathcal{C}\}$.

They are random because they take on random values based on the outcome of an experiment.



Example

A fair die is thrown twice. The sample points are $(1, 1), (1, 2), \dots, (6, 6)$. There are 36 sample points. Let us assign the same probability $1/36$ for each of these points. Suppose we are interested in the sum of the numbers of each outcome. Then we can define a random variable $X = i + j$ associated with the outcome (i, j) .

List the possible values of X and find the probability of each value.

Example

Value of X	Probability
2	$1/36$
3	$2/36$
4	$3/36$
5	$4/36$
6	$5/36$
7	$6/36$
8	$5/36$
9	$4/36$
10	$3/36$
11	$2/36$
12	$1/36$

Types of Random Variables

In this course, we will introduce two different types of random variables:

- (i) **Discrete** random variables
- (ii) **Continuous** random variables

We first will discuss the basic concepts of discrete random variables. We then will focus on continuous random variables.

Section 6

Discrete Random Variables

Discrete Random Variables

Definition 24

A random variable is said to be **discrete** if it can assume only a finite or countably infinite number of distinct values.

Example: If X represents the outcome of a six-sided die, it can only take on values 1, 2, 3, 4, 5, and 6.

Probability Distributions

- Capital letters denote random variables, typically X , Y , and Z and small letters denote the value of a random variable, typically x , y , and z . They are the outcomes of the experiment.
- We write $P(X = x)$ which in words is "the probability that random variable X takes on the value x ".
- Every random variable is associated with a **probability distribution** that specifies the possible random variable values and the probability each value will occur.

Probability Distributions

Definition 25

The **probability distribution** for a discrete random variable X can be represented by a formula, table, or a graph that provides $P(X = x)$ for all $x \in \mathcal{D}$.

- Note that $P(X = x) \geq 0$ for all $x \in \mathcal{D}$, but the probability distribution only assigns nonzero probabilities to a countable number of x values.
- It is convention that any value x not explicitly assigned a positive probability is understood to be zero, i.e., $P(X = x) = 0$.

Example

- Basically, the probability distribution of the random variable X tells us how the total probability of 1 is "distributed" among all the possible values that X can have.
- Example:** Let the random variable X be the sum of a pair of dice: $P(X = x) = \frac{6-|x-7|}{36}$, for $x = 2, 3, \dots, 12$, and it's zero, otherwise.

The table is the probability distribution of x .

x	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Probability Mass Functions

For discrete random variables, the probability distribution is called the **probability mass function** (**pmf**) and is denoted by $p_X(x) = P(X = x)$ for $x \in \mathcal{D}$.

Note that, for simplicity, we often remove the subscript X in $p_X(x)$ and just denote a pmf by $p(x)$.

Theorem 26

For any discrete probability distribution, the following must be true:

- ❶ $0 \leq p(x) \leq 1$ for all x .
- ❷ $\sum_x p(x) = 1$ where the summation is over all values of x with nonzero probability.

Note that these are just the probability axioms applied to random variables.

The induced probability distribution, $p_X(\cdot)$ of X is

$$P_X(D) = \sum_{d_i \in D} p(d_i), \quad D \subset \mathcal{D}.$$

Example

A recruiting manager has six internship applications, three men and three women. He has to choose two for a special project. So as not to play favorites, he decides to select the two at random. Let X denote the number of women selected. Find the probability distribution for X .

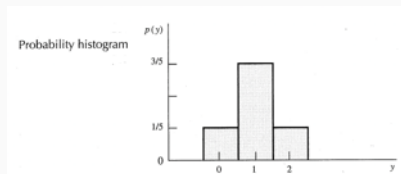
Solution: We can represent the result as a formula:

$$p(x) = \frac{\binom{3}{x} \binom{3}{2-x}}{\binom{6}{2}}, \quad x = 0, 1, 2.$$

Example

The result can also be summarized by a table or a histogram.

Probability distribution	
x	$p(x)$
0	$1/5$
1	$3/5$
2	$1/5$



Cumulative Distribution Functions

Definition 27

Let X denote any random variable. The **Cumulative Distribution Function** (cdf) of X , denoted by $F_X(x)$, is defined by

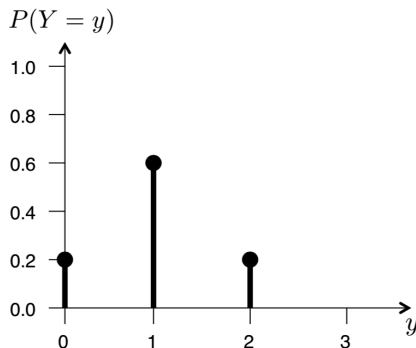
$$F_X(x) = P(X \leq x), \quad \text{for } -\infty < x < \infty.$$

Note that, for simplicity, we often remove the subscript X in $F_X(x)$ and just denote a cdf by $F(x)$.

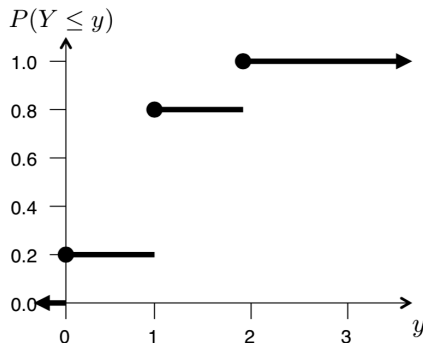
Example: pmf vs cdf

In the example,

Probability Mass Function



Cumulative Distribution Function



{Properties of cdf for a Discrete Random Variable}

Theorem 28

If $F(x)$ is a cdf, then

$$\textcircled{1} \quad F(-\infty) = \lim_{x \rightarrow -\infty} P(X \leq x) = \lim_{x \rightarrow -\infty} F(x) = 0$$

$$\textcircled{2} \quad F(\infty) = \lim_{x \rightarrow \infty} P(X \leq x) = \lim_{x \rightarrow \infty} F(x) = 1$$

$\textcircled{3} \quad F(x)$ is a right-continuous non-decreasing function of x .

Non-decreasing means that if x_1 and x_2 are any values such that $x_1 < x_2$ then $F(x_1) \leq F(x_2)$.

Some Notes

- Note that this is the same definition for both discrete and continuous random variables.
- But there is a difference between of the cdf of discrete and continuous random variables.
- The cdf for a discrete random variables is a step function. They jump at each x value at which there is positive probability. Otherwise, in between, the function is flat.
- This occurs because the cdf only increases at finite or countable number of points with positive probability.
- The function is always a nondecreasing function.

Transformations

A problem often encountered in statistics is the following. We have a random variable X and we know its distribution. We are interested, though, in a random variable Y which is some transformation of X , say, $Y = g(X)$. In particular, we want to determine the distribution of Y .

Assume X is discrete with support S_X . Then the support of Y is $S_Y = \{g(x) : x \in S_X\}$. If g is one-to-one. Then, the pmf of Y is obtained as

$$p_Y(y) = P(Y = y) = P[g(X) = y] = P[X = g^{-1}(y)] = p_X(g^{-1}(y)).$$

When the transformation, $g(x)$, is not one-to-one. Instead of developing an overall rule, for most applications with discrete random variables, the pmf of Y can be obtained in a straightforward manner.

Examples

- **Example 1:** Let X have a pmf $p_X(x) = \frac{1}{3}$ for $x = 1, 2, 3$ and zero elsewhere. Find the pmf of $Y = 2X + 1$.

Solution: Since $g(x) = 2x + 1$ is a one-to-one function, then for $y = 3, 5, 7$

$$p_Y(y) = P(Y = y) = P(2X + 1 = y) = P(X = \frac{y-1}{2}) = p_X(\frac{y-1}{2}) = \frac{1}{3}.$$

- **Example 2:** Let X have the pmf $p_X(x) = \frac{1}{3}$, for $x = -1, 0, 1$ and zero elsewhere. Find the pmf of $Y = X^2$.

Solution: Here $g(x) = x^2$ does not define a one-to-one transformation. But, we see that $S_Y = \{0, 1\}$ and

$$p_Y(1) = P(Y = 1) = P(X^2 = 1) = P(X = 1) + P(X = -1) = \frac{2}{3}$$

and

$$p_Y(0) = P(Y = 0) = P(X^2 = 0) = P(X = 0) = \frac{1}{3}$$

Section 7

Continuous Random Variables

Continuous Random Variables

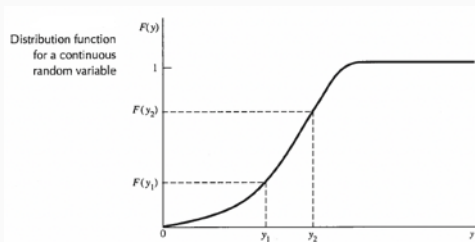
- Continuous random variables can take on any value within an interval. i.e., they have an uncountably infinite number of values they can take on.
- Recall that discrete random variables can take on either a finite or countably infinite number of values.
- **Example:** With a perfectly precise measuring instrument, height is continuous.

Continuous Random Variables

Definition 29

A random variable X with the cdf $F(x)$ is said to be **continuous** if $F(x)$ is continuous for $-\infty < x < \infty$.

What this means is that the distribution function for continuous random variables is a smooth function. The plot below is an example.



Probability Density Function (pdf)

Definition 30

Let $F(x)$ be the distribution function for a continuous random variable X . Then, the probability density function (pdf) $f(x)$ is given by

$$f(x) = \frac{dF(x)}{dx} = F'(x)$$

It follows from the above definition that

$$F(x) = \int_{-\infty}^x f(t)dt$$

Continuous Random Variables

An important mathematical distinction with continuous random variables is that.

- For a continuous random variable, the probability that any particular value x occurs is always zero. i.e., $P(X = x) = 0$ for every x .
- This requires both a change in how we think about continuous random variables and a change in notation.
- But while the differences are important, the intuition that we have developed about random variables will carry over.

Notation

- The notation $f(x)$ is shorthand notation for the pdf of random variable X evaluated at x . If we want to be explicit, we can write $f_X(x)$.
- With one random variable, it is usually clear without the subscript, but with two or more it can be confusing.
- For example, random variable X has the pdf $f_X(\cdot)$ and random variable Y has the pdf $f_Y(\cdot)$.
- Also, sometimes different letters used to distinguish them. For example, random variable X has the pdf $f(\cdot)$ and random variable Y has the pdf $g(\cdot)$.

Properties of Probability Density Functions

Theorem 31

If $f(x)$ is a pdf for a continuous random variable, then

- ❶ $f(x) \geq 0$ for all $x \in \mathcal{D}$.
- ❷ $\int_{-\infty}^{\infty} f(x)dx = 1$.

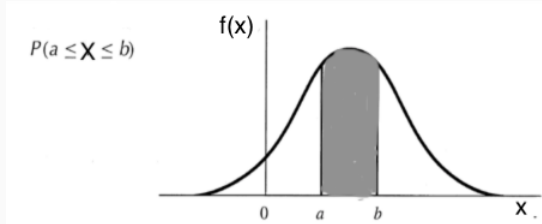
That is, pdfs are always non-negative and they must integrate to 1. Functions that do not have these properties cannot be density functions.

{Calculating Probabilities for Continuous Random Variables}

The induced probability distribution of X , $P_X(\cdot)$, is defined as

$$P_X[(a, b)] = P(a \leq X \leq b) = P[\{c \in \mathcal{C} : a < X(c) < b\}] = \int_a^b f(x)dx.$$

- Graphically: The area under the curve between a and b .



Remark

- Remember that if X is continuous, then

$$P(X = a) = P(X = b) = 0$$

- Thus,

$$P(a \leq X \leq b) = P(a < X < b)$$

- That is, whether the endpoints are included in the integration or not does not matter for continuous random variables.
- This is not true for discrete random variables.

Example

Given $f(x) = cx^2$, $0 \leq x \leq 2$, find c so that $f(x)$ is a valid pdf. Also, find $P(1 < X < 2)$.

Solution: From the properties of density functions

$$\int_0^2 cx^2 dy = 1$$

Hence, $c \left[\frac{x^3}{3} \Big|_0^2 \right] = 1$ which yields $c = \frac{3}{8}$.

Also,

$$P(1 < X < 2) = \int_1^2 \frac{3}{8}x^2 dx = \frac{7}{8}$$

Quantiles and Percentiles

Definition 32

Let $0 < p < 1$. The **quantile** of order p of the distribution of a random variable X is a value ξ_p such that $P(X < \xi_p) \leq p$ and $P(X \leq \xi_p) \geq p$. It is also known as the $(100p)$ th **percentile** of X .

Note that $\xi_{0.5}$ is the **median** that $P(X < \xi_{0.5}) \leq 0.5$ and $P(X \leq \xi_{0.5}) \geq 0.5$. The median is also called the second quantile or q_2 .

The difference $IQR = q_3 - q_1$ is called the **interquartile range** of X , where q_1 and q_3 represent the 25th and 75th percentiles of X , respectively. IQR is used as a measure of **spread** or **dispersion** of the distribution of X .

Quantiles need not be unique even for continuous random variables with pdfs.

Examples

- **Example 1:** Find the 0.20 quantile (20th percentile) of the distribution that has pdf $f(x) = 4x^3$, for $0 < x < 1$, and zero elsewhere.
- **Example 2:** Find the median of random variable X that has the following cdf.

$$F(x) = \begin{cases} 0, & x < 2 \\ 1/8, & 2 \leq x < 2.5 \\ 3/16, & 2.5 \leq x < 4 \\ 1/2, & 4 \leq x < 5.5 \\ 5/8, & 5.5 \leq x < 6, \\ 11/16, & 6 \leq x < 7, \\ 1, & x \geq 7. \end{cases}$$

Transformations: The cdf Technique

Let X be a continuous random variable with a known pdf f_X . As in the discrete case, we are often interested in the distribution of a random variable Y which is some transformation of X , say, $Y = g(X)$.

Often we can obtain the pdf of Y by directly obtaining its cdf of $Y = g(X)$.

Example: Let X be a random variable with the pdf $f_X(x) = \frac{1}{2}$ for $-1 < x < 1$. Find the pdf of $Y = X^2$.

Solution: The cdf of Y is obtained by

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y}) \end{aligned}$$

Taking the derivative from both sides, the pdf of Y becomes

$$f_Y(y) = \frac{dF(y)}{dy} = \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) + \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}) = \frac{1}{2\sqrt{y}} \times \frac{1}{2} + \frac{1}{2\sqrt{y}} \times \frac{1}{2} = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1.$$

Transformations: Jacobian Method

When the transformation $Y = g(X)$ is one-to-one, the following theorem can be used to find the pdf of Y .

Theorem 33

Let X be a continuous random variable with pdf $f_X(x)$ and support S_X . Let $Y = g(X)$, where $g(x)$ is a one-to-one differentiable function, on S_X . Denote the inverse of g by $x = g^{-1}(y)$ and let $dx/dy = d[g^{-1}(y)]/dy$. Then the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \quad \text{for } y \in S_Y$$

where the support of Y is the set $S_Y = \{y = g(x) : x \in S_X\}$.

We refer to $dx/dy = d[g^{-1}(y)]/dy$ as the **Jacobian** (denoted by J) of the transformation.

Examples

Example 1: Let X be a random variable with the pdf $f_X(x) = 2x$ for $0 < x < 1$. Find the pdf of $Y = X^2$.

Solution: Because $g(x) = x^2$, g is a one-to-one differentiable function on $(0, 1)$ and $g^{-1}(y) = \sqrt{y}$. Further, $\frac{dx}{dy} = \frac{d[g^{-1}(y)]}{dy} = \frac{1}{2\sqrt{y}}$. By the Theorem, we have

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right| = 2\sqrt{y} \frac{1}{2\sqrt{y}} = 1, \quad \text{for } y \in (0, 1)$$

Example 2: ([Exercise](#)) Let X be a random variable with the pdf $f_X(x) = 2(1 - x)$ for $0 < x < 1$. Find the pdf of $Y = -\ln(1 - X)$.

Section 8

Expectation of a Random Variable

{Expectation of a Random Variable}

Definition 34

Let X be a random variable. If X is a discrete random variable with pmf $p(x)$ and $\sum_x |x| p(x) < \infty$, then the expectation of X is

$$E(X) = \sum_x xp(x)$$

If X is a continuous random variable with pdf $f(x)$ and $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$, then the expectation of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

We often use μ as shorthand to denote the expected value. i.e, $\mu = E(X)$.

Sometime, the expectation $E(X)$ is called the **expected value** of X , or the **mean** of X .

Example

Back to the Example of choosing 2 out of 6 applications which consists of 3 men and 3 women. Here X is the number of female applications in the sample. The probability mass function is as follows. Find the expectation of X .

Probability distribution	
x	$p(x)$
0	$1/5$
1	$3/5$
2	$1/5$

$$\text{Then } \mu = E(X) = \sum_{x=0}^2 xp(x) = 0 \times \frac{1}{5} + 1 \times \frac{3}{5} + 2 \times \frac{1}{5} = 1$$

Example 2

Let X is random variable with the pdf $f(x) = \frac{x+2}{18}$ for $-2 < x < 4$, and zero elsewhere. Find $E(X)$.

Solution: By the definition,

$$E(X) = \int_{-2}^4 x \frac{x+2}{18} dx = \frac{1}{18} \left\{ \frac{x^3}{3} \Big|_{-2}^4 + x^2 \Big|_{-2}^4 \right\} = \frac{1}{18} \left(\frac{72}{3} + 12 \right) = 2$$

{Expectation of a Function of a Random Variable}

Theorem 35

Let X be a random variable, and let $Y = g(X)$ for some function g .

- Suppose X is discrete with pmf $p_X(x)$ and the support of X is denoted by S_X . If $\sum_{x \in S_X} |g(x)| p_X(x) < \infty$, then the expectation of Y exists and it is given by

$$E(Y) = \sum_{x \in S_X} g(x) p_X(x).$$

- Suppose X is continuous with pdf $f_X(x)$. If $\int_{-\infty}^{\infty} |g(x)| f_X(x) dx < \infty$, then the expectation of Y exists and it is given by

$$E(Y) = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

Examples

Example 1: Let X be a discrete random variable with pmf $p(x) = \left(\frac{1}{2}\right)^{x+1}$, for $x = 0, 1, 2, \dots$ and $Y = e^{-X}$. Find $E(Y)$.

Solution: By Theorem 35 and the formula for geometric series, we have

$$E(Y) = E(e^{-X}) = \sum_{x=0}^{\infty} e^{-x} \left(\frac{1}{2}\right)^{x+1} = \frac{1}{2} \sum_{x=0}^{\infty} \left(\frac{1}{2e}\right)^x = \frac{e}{2e-1}$$

Example 2: Let X be a continuous random variable with pdf $f(x) = 2x$, for $0 < x < 1$ and $Y = \frac{1}{1+X}$. Find $E(Y)$.

Solution: By Theorem 35 and the integration technique, we have

$$E(Y) = E\left(\frac{1}{1+X}\right) = \int_0^1 \frac{2x}{1+x} dx = \int_0^1 \frac{2x+2-2}{1+x} dx = [2x - 2\ln(1+x)]_0^1 = 2(1 - \ln 2)$$

Expectation: Linearity

Theorem 36

Let $g_1(X)$ and $g_2(X)$ be functions of a random variable X . Suppose the expectations of $g_1(X)$ and $g_2(X)$ exist. Then for any constants k_1 and k_2 , the expectation of $k_1g_1(X) + k_2g_2(X)$ exists and it is given by

$$E[k_1g_1(X) + k_2g_2(X)] = k_1E[g_1(X)] + k_2E[g_2(X)]$$

Proof: In class.

Example

Let X have the pmf $p(x) = \frac{x}{6}$ for $x = 1, 2, 3$ and 0 elsewhere. Find $E(6X^3 + X)$.

Solution: From the theorem, $E(6X^3 + X) = 6E(X^3) + E(X)$. First note that

$$E(X) = \sum_{x=1}^3 xp(x) = \sum_{x=1}^3 \frac{x^2}{6} = \frac{14}{6}$$

and

$$E(X^3) = \sum_{x=1}^3 x^3p(x) = \sum_{x=1}^3 \frac{x^4}{6} = \frac{98}{6}$$

$$\text{Hence, } E(6X^3 + X) = 6 \times \frac{98}{6} + \frac{14}{6} = \frac{602}{6} = \frac{301}{3}$$

Section 9

Some Special Expectations

Variance of a Random Variable

Definition 37

For a random variable X with mean $E(X) = \mu$, the variance of X is defined as the expected value of $(X - \mu)^2$. That is,

$$\text{Var}(X) = E[(X - \mu)^2]$$

- Variance is literally the average squared deviation of X from its mean.
- It is a measure of variability of a random variable. The bigger the variance, the more "spread out" (around the mean) the values that the random variable can take on.
- It is easy to show that

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$$

Proof: In class.

Standard Deviation of a Random Variable

- The **standard deviation** of X is the positive square root of $Var(X)$.
- Conventional notation often denotes the variance as $Var(X) = \sigma^2$. So, the standard deviation is often denoted by σ .
- Note that the scale of variance is square of that of mean. In comparison, the standard deviation is in the same units as the mean. This often makes the standard deviation easier to understand and interpret.

Example

Consider two users. One receives either 8 or 12 e-mail messages per day, with a 50-50% chance of each. The other receives either 0 or 20 emails, also with a 50-50% chance. Find the expectation, variance, and standard deviation of these two random variables.

Solution: Define X the number of emails received by user one

$$E(X) = 8 \times \frac{1}{2} + 12 \times \frac{1}{2} = 10, \quad E(X^2) = 8^2 \times \frac{1}{2} + 12^2 \times \frac{1}{2} = 104$$

Hence, the variance becomes $\sigma_X^2 = E(X^2) - (E[X])^2 = 104 - 100 = 4$ and the standard deviation is $\sigma_X = \sqrt{4} = 2$.

Now, let Y be the number of emails received by user 2. Then, similarly, we see that $E(Y) = 10$, $\sigma_Y^2 = 100$, and thus $\sigma_Y = 10$.

Variance of a Random Variable with Constants

Theorem 38

Let X be a random variable with finite mean μ and variance σ^2 . Then for all constants a and b ,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Proof: In class.

Moment Generating Functions

Definition 39

Let X be a random variable such that for some $h > 0$, $E(e^{tX})$ exists for $-h < t < h$. The moment generating function (mgf) of X is defined to be the function $M(t) = E(e^{tX})$, for $-h < t < h$.

Moment generating functions can uniquely identify distributions. This means that if we let X and Y be two random variables with mgfs, and let X and Y have the same distribution, i.e, $F_X(z) = F_Y(z)$ for all z . Then certainly $M_X(t) = M_Y(t)$ in a neighborhood of 0. Note that the converse of this statement is true too.

Moment Generating Functions: Applications

The moment-generating function possesses two important applications.

- If we can find $E(e^{tX})$, we can find $E(X^k)$. We call $E(X^k)$ the k th **moment of the distribution**, or the k th moment of X .
- The second application of a moment-generating function is to prove that a random variable possesses a particular probability distribution $p(x)$. If $M(t)$ exists for a probability distribution $p(x)$, it is unique.

Theorem 40

If $M(t)$ exists, then for any positive integer k

$$\left. \frac{d^k M(t)}{dt^k} \right|_{t=0} = M^{(k)}(0) = E(X^k)$$

In other words, if you find the k th derivative of $M(t)$ with respect to t and then set $t = 0$, the result will be $E(X^k)$.

Example

Let $M(t) = \frac{1}{6}e^t + \frac{2}{6}e^{2t} + \frac{3}{6}e^{3t}$. Find $E(X)$, $Var(X)$, and the distribution of X .

- **Solution:** $E(X) = \frac{dM(t)}{dt}\big|_{t=0} = 14/6$
- **Solution:** First, $E(X^2) = \frac{d^2M(t)}{dt^2}\big|_{t=0} = \frac{36}{6} = 6$. Hence,
 $Var(X) = 6 - (14/6)^2 = 0.556$
- **Solution:** From the definition of $M(t)$, i.e $M(t) = E(e^{tX})$, it is easy to see that

x	1	2	3
p(x)	1/6	2/6	3/6

Section 10

Important Inequalities

Markov's Inequality

Theorem 41

Let $u(X)$ be a nonnegative function of the random variable X . If $E[u(X)]$ exists, then for every positive constant c ,

$$P(u(X) \geq c) \leq \frac{E[u(X)]}{c}.$$

Proof: In class.

Chebyshev's Inequality

The preceding theorem is a generalization of an inequality that is often called **Chebyshev's Inequality**.

Theorem 42

Let X be a random variable with finite variance σ^2 . Then for every $k > 0$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

or equivalently,

$$P(|X - \mu| < k\sigma) > 1 - \frac{1}{k^2}$$

Proof: In class.

Example

Let X have the $f(x) = \frac{1}{2\sqrt{3}}$ for $-\sqrt{3} < x < \sqrt{3}$, find $P(|X - \mu| \geq k\sigma)$ for $k = \frac{3}{2}$ and compare it by the upper bound provided by Chebyshev's inequality.

Solution: We can show that $\mu = 0$ and $\sigma^2 = 1$, hence

$$P(|X - \mu| \geq k\sigma) = P(|X| \geq \frac{3}{2}) = 1 - \int_{-3/2}^{3/2} \frac{1}{2\sqrt{3}} dx = 1 - \frac{\sqrt{3}}{2} \approx 0.134.$$

By Chebyshev's inequality, this probability has the upper bound $\frac{1}{k^2} = \frac{4}{9} \approx 0.444$.

Hence the exact probability in this case is considerably less than the upper bound provided by Chebyshev's inequality.

Convex Functions

Definition 43

A function ϕ defined on an interval (a, b) , $-\infty \leq a < b \leq \infty$, is said to be **convex** function if for all x, y in (a, b) and for all $0 < \gamma < 1$,

$$\phi[\gamma x + (1 - \gamma)y] \leq \gamma\phi(x) + (1 - \gamma)\phi(y).$$

If the inequality is strict, we say ϕ is **strictly convex**.

Jensen's Inequality

Theorem 44

If ϕ is convex on an open interval I and X is a random variable whose support is contained in I and has finite expectation, then

$$\phi[E(X)] \leq E[\phi(X)]$$

If ϕ is strictly convex, then the inequality is strict unless X is a constant random variable.

Examples

Exmaple 1: Let X be a positive random variable, argue that

$$E(1/X) \geq 1/E(X)$$

Solution: Since $\phi(x) = 1/x$ is a convex function and the support of X is contained in $(0, \infty)$, the result follows from Jensen's inequality.

Exmaple 2: Let X be a positive random variable, argue that

$$E[-\log(X)] \geq -\log E[X]$$

Solution: Since $\phi(x) = -\log(x)$ is a convex function and the support of X is contained in $(0, \infty)$, the result follows from Jensen's inequality.

What We Have Learned

- Introduction
- Sets
- The Probability Set Function
- Conditional Probability and Independence
- Random Variables
- Discrete Random Variables
- Continuous Random Variables
- Expectation of a Random Variable
- Some Special Expectations
- Important Inequalities

The End.